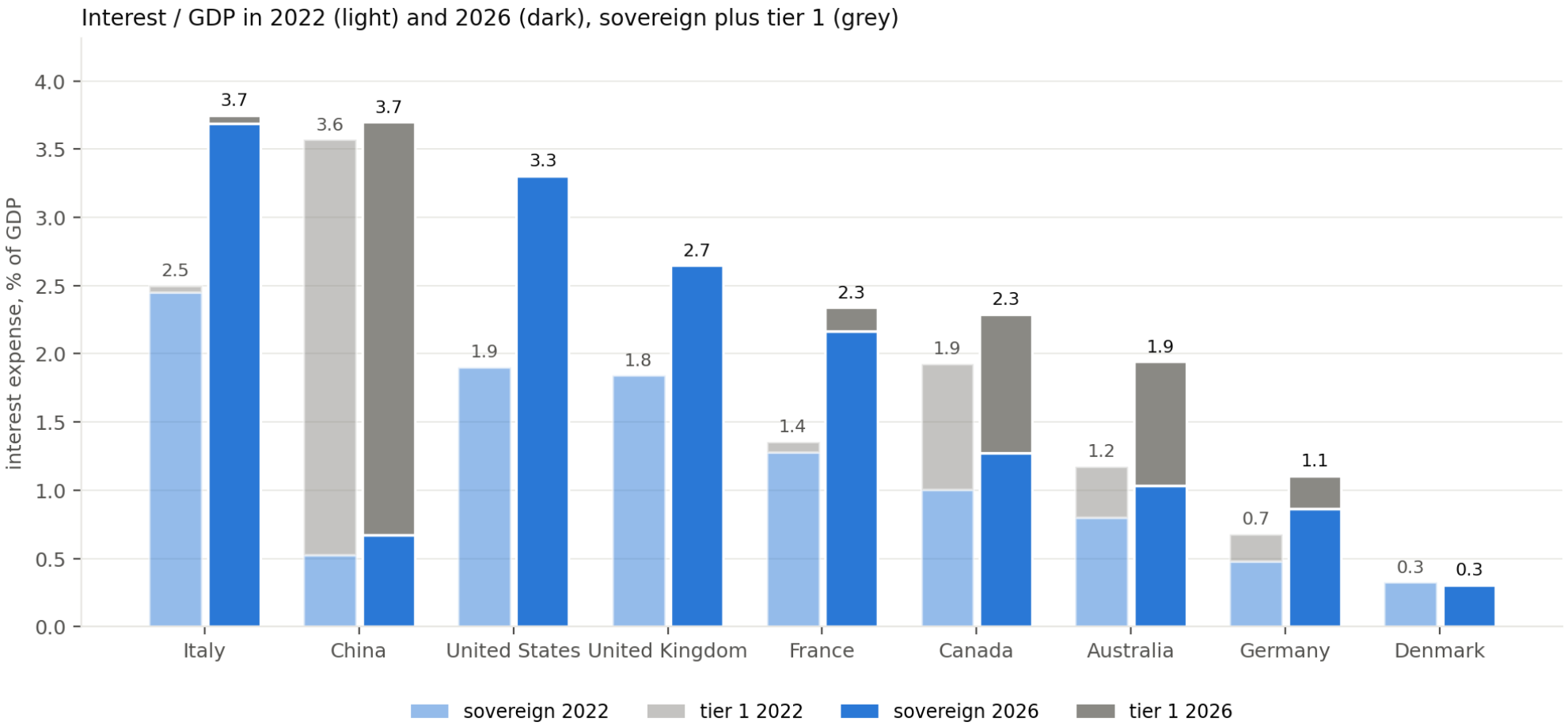
Where the interest burden stands today: 2022 against 2026

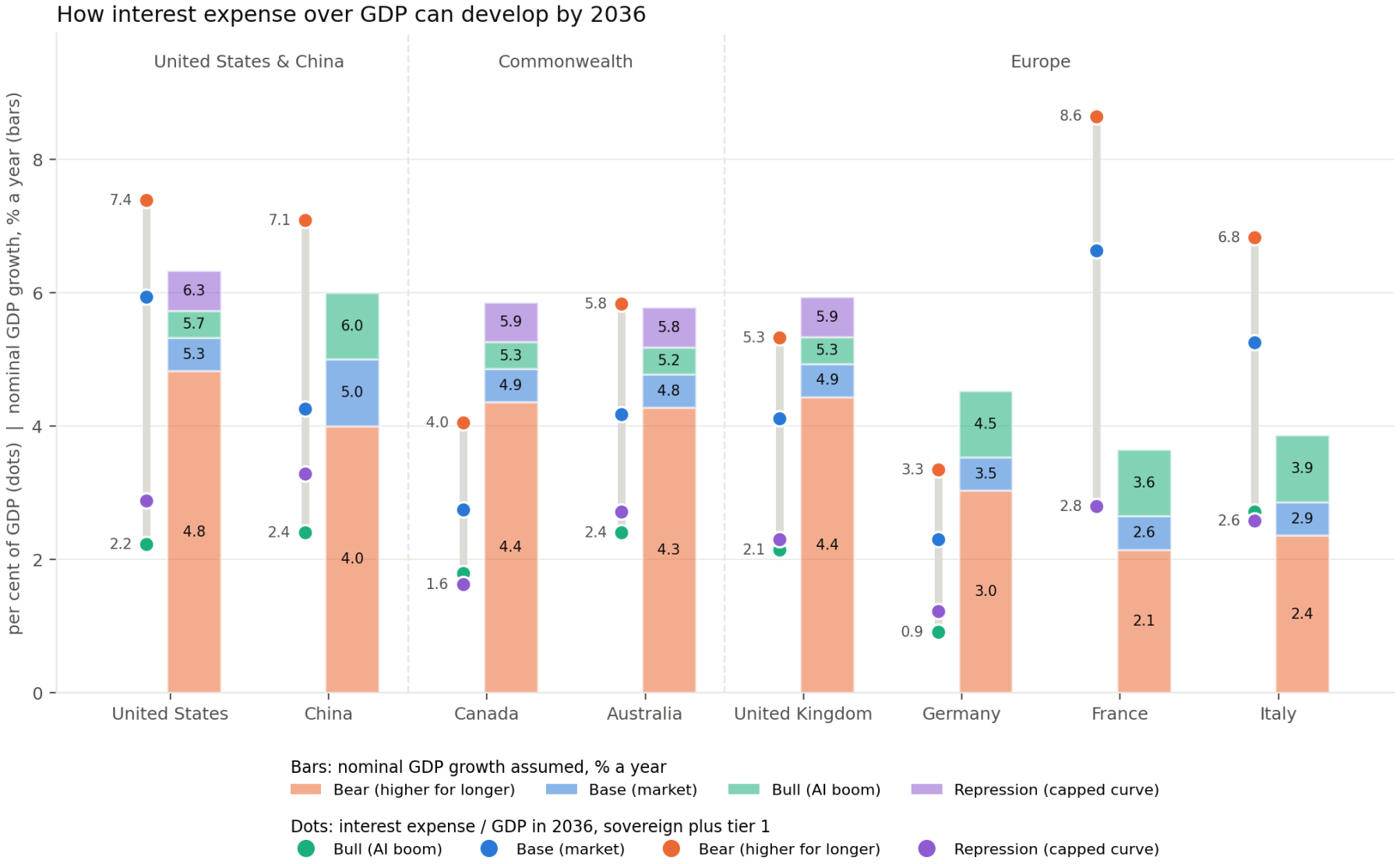
Bars: interest expense as a share of GDP in 2022 (light) and 2026 (dark); the coloured segment is the sovereign, the grey segment the fiscal tier-1 blocks (provinces, states, local government, Länder, French and Italian agencies). 2026 is the model's starting point, identical to the base-case bars elsewhere in this deck. Countries are ordered by the 2026 total.



2022 is rebuilt on the model's definitions from published figures: each sovereign's end-2022 marketable stock and average coupon or cash cost (US Treasury, Finance Canada, DMO/ONS, AFT and Cour des comptes, MEF, Finanzagentur, Danmarks Nationalbank, AOFM, Chinese MoF), the tier-1 blocks' 2022 stocks from the project's tier-1 files at 2022 effective rates (published for Ontario, NSW, CADES and Unédic; estimated for the rest), and World Bank 2022 nominal GDP. The UK rate is coupon interest net of the RPI uplift on index-linked gilts. Full sources per country are in INTEREST_THEN_VS_NOW.csv.

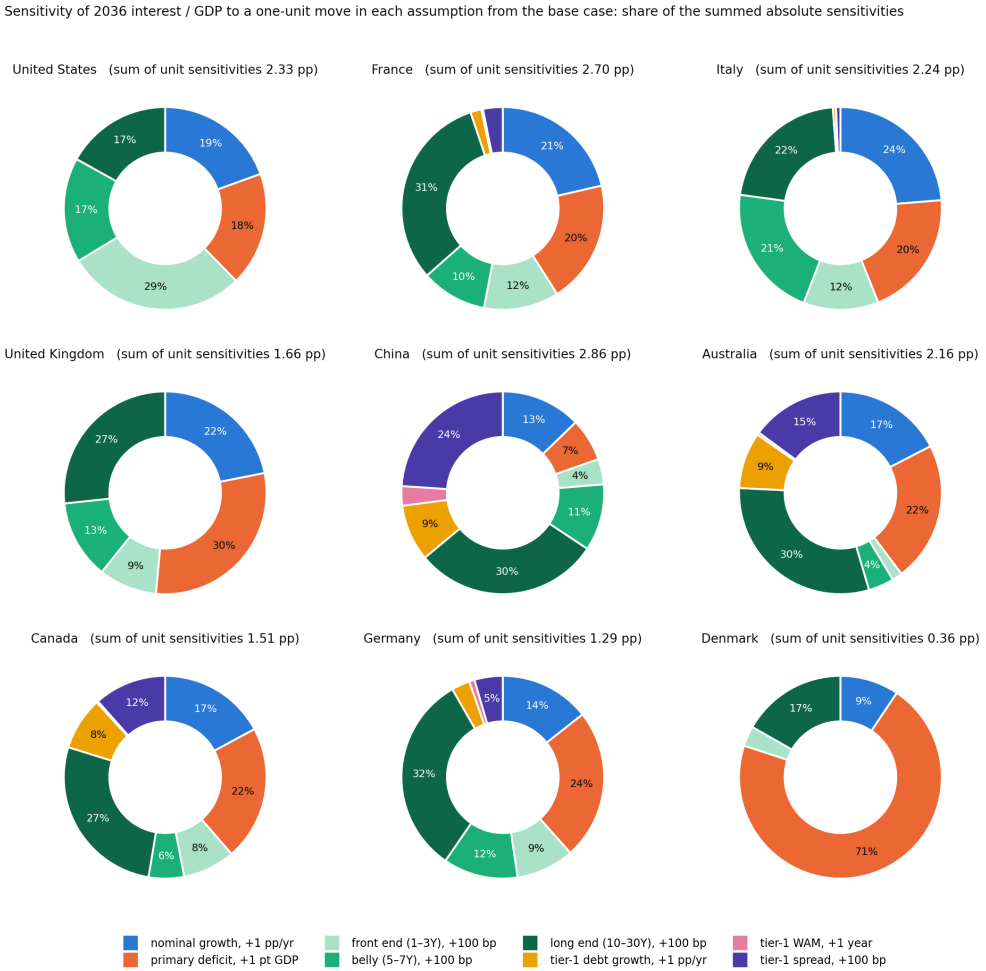
How interest expense over GDP can develop by 2036

For each country the dots are interest expense as a share of GDP in 2036 under the four cases, sovereign plus tier 1 (green = bull, blue = base, orange = bear, violet = repression; for France, Italy and Germany the repression case is an ECB-bloc decision rather than a national choice). The stacked bar beside them shows the nominal GDP growth assumed in each case, % a year, built up from the lowest case: each segment is the extra growth that case adds over the one below it, so the top of each segment is that case's growth rate (the bull is its 2027–36 average). Growth and interest share the same percentage axis. Countries are grouped by region; Denmark is left out of this chart because its bill is near zero in every case.



Which assumptions the ratio is most sensitive to

A like-for-like sensitivity rather than a decomposition of the scenarios: each assumption is moved by one unit from the base case with everything else at base, and the change in 2036 interest/GDP (sovereign plus tier 1) is recorded as the average of the up and down moves. The units are the conventional ones: +1 pp a year of nominal growth, +1 pt of GDP of primary deficit held every year, +100 bp on one segment of the curve at a time from 2027 (front end = the 1- to 3-year, belly = 5- and 7-year, long end = 10- to 30-year, each built from bumps at the tenors the sovereign issues at that fade linearly to the neighbouring tenors, so tier-1 paper priced at odd maturities moves too; the three segments add up to a parallel shift), +1 pp a year of tier-1 stock growth, +1 year of tier-1 WAM, +100 bp of tier-1 spread. Each slice is that lever's share of the summed absolute sensitivities, so the picture does not depend on how large the bull and bear settings happen to be. A segment's slice is the product of how much the country issues there and how long that paper then sits on the books at the new rate.



Assumptions behind each case

One page per case. Growth is nominal GDP growth, % a year, with its split into real growth and inflation in brackets (an arrow marks a change over the horizon). Base real growth is the average of the 2016–25 real GDP CAGR and 2025 (World Bank, constant prices); base inflation is nominal less real, i.e. the GDP deflator. The bull adds +1 real with inflation unchanged in the boom, then +1.5 real against –1.5 inflation (US, UK, Canada, Australia) or –0.5 (euro members, Denmark); the bear is –1.5 real and +1 inflation; repression is +1 inflation with real growth at base. Primary balance is % of GDP with + = deficit, shown at 2027, 2030 and 2036; where a country has tier 1 the sovereign runs on the central balance (general government less the sub-national sector's own balance) so sub-national borrowing is counted once. The curve is the new-issue rate at 1, 10 and 30 years in 2028 and in 2036; every other tenor moves with the nearest set points. Tier 1 is summarised across the fiscal blocks: stock as % of GDP, stock growth in the first year, debt-weighted WAM and spread over the sovereign curve (an arrow marks a change over the horizon). * = ECB-bloc case, not a national choice.

Bull: AI boom, then disinflation

AI boom, then disinflation, in two phases. 2027–30: a capex boom financed on market forwards, nominal growth 1 pt above base with inflation still 3–4%, and a revenue harvest that takes the primary balance to a surplus of at least 1% of GDP by 2030 (1.5 pt better where already in surplus). From 2031 the buildout is done, inflation falls and central banks cut: the curve glides over 2031–32 to a low terminal level and holds; growth falls back to base for the US, UK, Canada and Australia and stays base +1 for the euro members, Denmark and China; the surplus is held. Tier 1 keeps its middle growth pace and moves to the longer WAM and tight spread as the cuts arrive. China's curve stays at market (the cap holds).

Country	Growth, %/yr nominal (real + inflation)	Primary balance, % GDP 2027 / 2030 / 2036	Curve 2028 1Y / 10Y / 30Y	Curve 2036 1Y / 10Y / 30Y	Tier 1 (fiscal blocks)
United States	6.3 (3.3 + 3.0) → 5.3 (3.8 + 1.5)	+3.6 / -1.0 / -1.0	4.6 / 5.1 / 5.4	2.0 / 3.0 / 3.5	—
France	3.6 (2.0 + 1.6)	+1.7 / -1.0 / -1.0	3.9 / 5.1 / 5.5	1.2 / 2.8 / 3.2	CADES, Unédic, ACOSS, 8% of GDP: growth +3.2%/yr; WAM 4 → 5y; spread +20 → 10 bp
Italy	3.9 (1.8 + 2.1)	-1.3 / -2.4 / -2.4	4.0 / 4.7 / 5.2	1.2 / 2.9 / 3.3	CDP, FS, 2% of GDP: growth +4.0%/yr; WAM 5 → 7y; spread +30 → 20 bp
United Kingdom	5.9 (2.3 + 3.6) → 4.9 (2.8 + 2.1)	+0.7 / -1.0 / -1.0	4.7 / 5.4 / 5.8	1.8 / 2.9 / 3.4	—
China	6.0 (6.2 + -0.2)	+2.8 / -1.0 / -1.0	1.2 / 1.7 / 2.1	1.5 / 1.8 / 2.1	local govt + LGFV, 80% of GDP: growth LGB +16%, LGFV -3%; WAM 6 → 8y; spread +88 → 52 bp
Australia	5.8 (2.8 + 3.0) → 4.8 (3.3 + 1.5)	+0.5 / -1.0 / -1.0	4.6 / 5.2 / 5.7	1.8 / 2.9 / 3.4	states, 23% of GDP: growth +10.7%/yr; WAM 7 → 9y; spread +51 → 41 bp
Canada	5.9 (2.9 + 2.9) → 4.9 (3.4 + 1.4)	-0.1 / -1.3 / -1.3	3.2 / 3.9 / 4.2	1.5 / 2.5 / 2.9	provinces, 30% of GDP: growth +4.8%/yr; WAM 11 → 14y; spread +70 → 47 bp
Germany	4.5 (1.4 + 3.1)	+1.5 / -1.0 / -1.0	2.9 / 3.4 / 3.7	1.0 / 1.8 / 2.2	Länder, 14% of GDP: growth +0.2%/yr; WAM 6 → 8y; spread +36 → 26 bp
Denmark	5.5 (3.5 + 2.0)	-0.9 / -2.0 / -2.0	2.7 / 3.3 / 3.6	0.8 / 1.6 / 2.0	—

Base: market forwards

Market pricing. New issuance is priced at the forward rates implied by today's curve with today's term premium; nominal growth is the average of the 2016–25 CAGR and the latest year; the primary balance is held at today's level; tier 1 grows at its middle historical pace with its current WAM and through-cycle spread. China's administered curve steepens the way a developed market's would (long end +0.75 to +1 pt).

Country	Growth, %/yr nominal (real + inflation)	Primary balance, % GDP 2027 / 2030 / 2036	Curve 2028 1Y / 10Y / 30Y	Curve 2036 1Y / 10Y / 30Y	Tier 1 (fiscal blocks)
United States	5.3 (2.3 + 3.0)	+5.2 / +5.2 / +5.2	4.6 / 5.1 / 5.4	4.5 / 5.2 / 5.4	—
France	2.6 (1.0 + 1.6)	+2.6 / +2.6 / +2.6	3.9 / 5.1 / 5.5	4.7 / 5.2 / 5.5	CADES, Unédic, ACOSS, 8% of GDP: growth +3.2%/yr; WAM 4y; spread +20 bp
Italy	2.9 (0.8 + 2.1)	-0.9 / -0.9 / -0.9	4.0 / 4.7 / 5.2	4.3 / 4.8 / 5.2	CDP, FS, 2% of GDP: growth +4.0%/yr; WAM 5y; spread +30 bp
United Kingdom	4.9 (1.3 + 3.6)	+1.2 / +1.2 / +1.2	4.7 / 5.4 / 5.8	4.6 / 5.4 / 5.5	—
China	5.0 (5.2 + -0.2)	+4.1 / +4.1 / +4.1	1.2 / 2.4 / 3.1	1.5 / 2.6 / 3.1	local govt + LGFV, 80% of GDP: growth LGB +16%, LGFV -3%; WAM 6y; spread +88 bp
Australia	4.8 (1.8 + 3.0)	+0.9 / +0.9 / +0.9	4.6 / 5.2 / 5.7	4.6 / 5.2 / 5.7	states, 23% of GDP: growth +10.7%/yr; WAM 7y; spread +51 bp
Canada	4.9 (1.9 + 2.9)	+0.2 / +0.2 / +0.2	3.2 / 3.9 / 4.2	3.3 / 3.8 / 4.2	provinces, 30% of GDP: growth +4.8%/yr; WAM 11y; spread +70 bp
Germany	3.5 (0.4 + 3.1)	+2.4 / +2.4 / +2.4	2.9 / 3.4 / 3.7	3.1 / 3.5 / 3.8	Länder, 14% of GDP: growth +0.2%/yr; WAM 6y; spread +36 bp
Denmark	4.5 (2.5 + 2.0)	-0.5 / -0.5 / -0.5	2.7 / 3.3 / 3.6	2.6 / 3.1 / 3.7	—

Bear: inflation persists, higher for longer

Inflation persists and rates stay higher for longer. No cuts: the front end sits 25 bp over market and the term premium rebuilds in proportion to the fiscal position (30-year +110 bp for the high-debt countries, +65 bp for the low-debt ones, France +50 and Italy +75 bp of spread from ten years out), reached by 2028 and held as a spread over market forwards. Nominal growth 0.5 pt below base, the primary deficit 1.5 pt of GDP worse by 2036, tier 1 at its fastest historical pace with the shorter WAM and stressed spread. China: the yield cap fails, a parallel +1.5 pt shift.

Country	Growth, %/yr nominal (real + inflation)	Primary balance, % GDP 2027 / 2030 / 2036	Curve 2028 1Y / 10Y / 30Y	Curve 2036 1Y / 10Y / 30Y	Tier 1 (fiscal blocks)
United States	4.8 (0.8 + 4.0)	+5.3 / +5.8 / +6.7	4.9 / 5.9 / 6.5	4.8 / 6.1 / 6.5	—
France	2.1 (-0.5 + 2.6)	+2.8 / +3.2 / +4.1	4.2 / 6.0 / 6.6	5.0 / 6.1 / 6.6	CADES, Unédic, ACOSS, 8% of GDP: growth +6.1%/yr; WAM 3y; spread +30 bp
Italy	2.4 (-0.7 + 3.1)	-0.7 / -0.3 / +0.6	4.3 / 5.5 / 6.3	4.7 / 5.6 / 6.3	CDP, FS, 2% of GDP: growth +7.1%/yr; WAM 4y; spread +40 bp
United Kingdom	4.4 (-0.2 + 4.6)	+1.4 / +1.8 / +2.7	5.0 / 6.2 / 6.9	4.9 / 6.2 / 6.6	—
China	4.0 (3.7 + 0.3)	+4.2 / +4.7 / +5.6	2.7 / 3.2 / 3.6	3.0 / 3.3 / 3.6	local govt + LGFV, 80% of GDP: growth LGB +16%, LGFV -3%; WAM 5y; spread +206 bp
Australia	4.3 (0.3 + 4.0)	+1.1 / +1.5 / +2.4	4.9 / 6.0 / 6.4	4.9 / 6.0 / 6.4	states, 23% of GDP: growth +11.9%/yr; WAM 5y; spread +90 bp
Canada	4.4 (0.4 + 3.9)	+0.4 / +0.8 / +1.7	3.5 / 4.7 / 4.9	3.6 / 4.6 / 4.9	provinces, 30% of GDP: growth +7.1%/yr; WAM 8y; spread +95 bp
Germany	3.0 (-1.1 + 4.1)	+2.5 / +3.0 / +3.9	3.2 / 4.2 / 4.4	3.4 / 4.2 / 4.4	Länder, 14% of GDP: growth +3.2%/yr; WAM 4y; spread +65 bp
Denmark	4.0 (1.0 + 3.0)	-0.4 / +0.1 / +1.0	3.0 / 4.0 / 4.3	2.9 / 3.9 / 4.3	—

Repression: inflation 3–4%, curve capped

Inflation of 3–4% with the curve engineered down. The full easing cycle is held for the decade (a negative real policy rate) and the long end is capped with the term premium near zero, so the curve is flat (10-year about 1.3 pt over the 1-year), reached by 2028 and held as a level. Nominal growth 1 pt above base (inflation, not real growth); the primary balance is held, because repression substitutes for consolidation; tier 1 at its middle pace with the longer WAM and tight spread as captive banks absorb it. For Germany, France and Italy this is an ECB-bloc case, not a national option; Australia gets a shallower cap; China already runs this curve. Bondholders pay: a 10-year near 3.3% against 3.5–4% inflation.

Country	Growth, %/yr nominal (real + inflation)	Primary balance, % GDP 2027 / 2030 / 2036	Curve 2028 1Y / 10Y / 30Y	Curve 2036 1Y / 10Y / 30Y	Tier 1 (fiscal blocks)
United States	6.3 (2.3 + 4.0)	+5.2 / +5.2 / +5.2	1.9 / 3.3 / 3.6	1.9 / 3.3 / 3.6	—
France*	3.6 (1.0 + 2.6)	+2.6 / +2.6 / +2.6	1.8 / 2.5 / 2.7	1.8 / 2.5 / 2.7	CADES, Unédic, ACOSS, 8% of GDP: growth +3.2%/yr; WAM 5y; spread +10 bp
Italy*	3.9 (0.8 + 3.1)	-0.9 / -0.9 / -0.9	1.9 / 2.7 / 2.9	1.9 / 2.7 / 2.9	CDP, FS, 2% of GDP: growth +4.0%/yr; WAM 7y; spread +20 bp
United Kingdom	5.9 (1.3 + 4.6)	+1.2 / +1.2 / +1.2	1.9 / 3.4 / 3.7	1.9 / 3.4 / 3.7	—
China	5.0 (5.2 + -0.2)	+4.1 / +4.1 / +4.1	1.2 / 1.7 / 2.1	1.5 / 1.8 / 2.1	local govt + LGFV, 80% of GDP: growth LGB +16%, LGFV -3%; WAM 8y; spread +52 bp
Australia	5.8 (1.8 + 4.0)	+0.9 / +0.9 / +0.9	2.0 / 3.7 / 4.1	2.0 / 3.7 / 4.1	states, 23% of GDP: growth +10.7%/yr; WAM 9y; spread +41 bp
Canada	5.9 (1.9 + 3.9)	+0.2 / +0.2 / +0.2	1.1 / 2.4 / 2.7	1.1 / 2.4 / 2.7	provinces, 30% of GDP: growth +4.8%/yr; WAM 14y; spread +47 bp
Germany*	4.5 (0.4 + 4.1)	+2.4 / +2.4 / +2.4	1.3 / 2.0 / 2.2	1.3 / 2.0 / 2.2	Länder, 14% of GDP: growth +0.2%/yr; WAM 8y; spread +26 bp
Denmark	5.5 (2.5 + 3.0)	-0.5 / -0.5 / -0.5	1.2 / 1.9 / 2.1	1.2 / 1.9 / 2.1	—

Sources

Every number in this deck traces to one of the inputs below. Model code: debt_model.py (cohort roll-forward engine, validated line by line against US_Debt_Model_v4.xlsx), build_inputs.py (turns the Bloomberg pulls into the input files), run_scenarios.py (scenario definitions), tier1_overlay.py (sub-sovereign blocks), scenario_lines.py and then_vs_now.py (these charts). Snapshot date 16 October 2026 unless stated.

Debt stock, coupons and maturity ladders (2026 and 2016 vintages)

- Bloomberg SRCH pulls of every live central-government bond and bill per country (data/bloomberg/grid_*.xlsx, not redistributed): amount outstanding, coupon, maturity, type. 2026 pulls dated 16 Oct 2026; 2016 pulls dated 30 Sep 2016 (US), 31 Mar 2016 (UK, Canada), 30 Jun 2016 (Australia) and end-2016 for the rest. Issue dates are not in the pull and are estimated from market conventions and the chronological order of CUSIP / SEDOL / ISIN codes, anchored on the 2016 pull.

- Issuance composition = lines opened in the last two years by original tenor (Australia: inferred from the maturity profile). Average cost of the stock = coupon-weighted on coupon debt plus bills at the 1-year rate. US pull excludes TIPS and FRNs; CB holdings ignored.

Yield curves and term premia

- On-the-run government yields, 16 Oct 2026 (data/bloomberg/grid_xzbzwrw2.xlsx) for the US, Canada, UK, Germany, France and Italy. Short end from public sources, mid-Oct 2026: policy rates (Fed 3.75–4.00, BoC 2.25, BoE 3.75, ECB deposit 2.50, Danmarks Nationalbank 2.10, RBA 4.60, PBoC 7-day reverse repo 1.40) and 1-year bills (US 1y CMT 16 Oct; Canada 1y bill 14 Oct; UK ~1y gilt 19 Oct; 12m Bubill 12 Oct; 12m BTF 19 Oct; 12m BOT mid-Oct; China 1y CGB 20 Oct).

- Curves not in the yields file: China from the ChinaBond CGB curve, 20 Oct 2026; Denmark 2y and 10y quotes of 19 Oct 2026 with 1y, 5y and 30y interpolated; Australia 3y/5y/10y of 15 Oct 2026, 30y = 10y plus the early-October 30–10 spread, 1y interpolated.

- 2016 spot curves: user-supplied table (overnight, 2y, 5y, 10y, long bond mapped to 30y; 1y interpolated).

- Term premia (5y and 10y, 2016 and 2026): user's OIS decomposition workbook for the US, Canada and UK. Euro members and Denmark built as EUR OIS term premium plus the Bund swap spread plus the 10-year sovereign spread versus Bunds (Eurostat convergence yields Dec 2016: OAT +50 bp, BTP +164 bp, DGB +14 bp; Oct 2026: OAT +150.8 bp, BTP +112.4 bp, DGB –17.6 bp). Australia set equal to Canada by instruction. China a judgment estimate anchored on the CGB-versus-policy-rate gap. (data/bloomberg/country_macro_comparison.xlsx, tabs Country Data, TP Build, Notes & Sources.)

Fiscal balances and GDP

- General-government fiscal balances: IMF World Economic Outlook, October 2026 (2026 values are projections; UK 2026 from the April 2026 Fiscal Monitor; US and UK 2016 via FRED). Primary deficit = IMF balance less the model's own interest. Where tier 1 exists the sovereign runs on a central balance = general government less the tier-1 sector's own 2026 balance (from the tier-1 files).

- Nominal GDP, local currency: World Bank WDI, GDP current LCU (NY.GDP.MKTP.CN), 2015, 2016, 2024 and 2025; 2026 extrapolated at the 2025 growth rate. Base nominal growth = average of the 2016–25 CAGR and 2025.

Tier-1 (sub-sovereign and quasi-sovereign) debt

- data/tier1/*.csv, built from issuer disclosures 2016–2026: Canadian provincial financing authorities (Ontario, Quebec, Alberta, BC: total debt, budget balance, WAM), Australian state treasury corporations (NSW TCorp, TCV, QTC, WATC), Chinese Ministry of Finance local-government bond data and LGFV bond buckets by rating, German Länder debt statistics (securities, Schuldschein and loans), French agencies (CADES, Unédic, ACOSS; SFIL/CAFFIL, Bpifrance, AFL, IDFM as exposure variants), Italian CDP, FS, regional sinkers and CDP postal savings. Spreads: through-cycle and current (Oct 2026) over the sovereign curve. Method notes in data/tier1/tier1_model_notes.pdf.

- LGFV bank and trust loans (~RMB 42tn) are hardcoded from the notes; no securities screen captures them.

- Starting effective interest rates (latest published interest paid / average debt): Ontario 3.3, Quebec 3.7, Alberta 3.1, BC 3.2; NSW 3.7, Victoria 4.6, Queensland 3.8, WA 3.3; MoF local-government bonds ~3.1; LGFV buckets ~3.6 / 4.3 / 5.2, loans ~4.5; NRW 1.8 and city-states 1.6; CADES 1.9, Unédic 1.4, ACOSS ~ESTR + 8 bp; SFIL/Bpifrance 2.4; CDP bonds and FS 3.0; CDP postal 2.8 all-in.

Scenario settings

• Bull terminal curves (2032), bear curves (2028), repression curves (2028), growth and deficit rules and the tier-1 bull / bear settings: specified by the author of this deck (run_scenarios.py, tier1_overlay.py). China's cases and the ECB-bloc flag for repression likewise.

The 2022-against-2026 chart

• US Treasury, Monthly Statement of the Public Debt and Average Interest Rates on marketable securities, 31 Dec 2022 (total marketable 2.32%)

• Department of Finance Canada, Debt Management Report 2022-23: market debt C$1,259.9bn, average rate 2.28%

• DMO Quarterly Review Jan-Mar 2023 (gilts £2,308bn uplifted nominal + T-bills £70bn); rate = ONS central-government debt interest 2022-23 (£111.6bn) less the RPI capital uplift on index-linked gilts (~2.6% of GDP, OBR) over the portfolio: estimate, about 2.0%

• AFT, dette negociable de l'Etat fin 2022 €2,278bn; Cour des comptes / Senat, charge de la dette 2022 €49.5bn of which indexation €15.53bn

• MEF, Rapporto sul debito pubblico 2022: titoli di Stato in circolazione €2,289.2bn; costo medio di cassa 2022 2.14%

• Finanzagentur, Kreditaufnahmebericht 2022: Umlaufvolumen €1,877.4bn, Eigenbestand €202.3bn; average interest rate 1.15% at end-2022 (IW Koeln)

• Danmarks Nationalbank, Central government borrowing and debt 2022: domestic debt DKK 628.9bn, interest on domestic debt DKK 9.3bn

• AOFM Annual Report 2021-22: AGS A$895bn at 30 June 2022, interest expense A$18.75bn

• Ministry of Finance, 2022 central final accounts: CGB balance RMB 25,869bn, debt interest expenditure RMB 652.4bn

• World Bank WDI, GDP current LCU (NY.GDP.MKTP.CN), 2022

• Tier-1 2022 stocks from the data/tier1 files; 2022 effective rates: provinces 3.30% (est.; Ontario 3.2% effective rate FY2022-23 (OFA)); semis 2.20% (est.; NSW TCorp average cost of debt 2.1% at 30 Jun 2022); LGB 3.40% (est.; 2022 LGB issuance averaged 3.02% (MoF), older vintages higher); LGFV AAA 4.00% (est.); LGFV AAplus 4.70% (est.); LGFV AA below 5.60% (est.); LGFV loans 5.00% (est.; LPR-based); Laender 1.30% (est.; Bundesbank Laenderfinanzen); CADES 1.29% (CADES average financing rate end-2022); UNEDIC 0.49% (Unedic cost of financing 2022); ACOSS 0.00% (est.; bill funding at zero / negative rates in 2022); core 2.50% (est.); LGFV loans 38,000 RMB bn (est.).

• 2026 is the model's own base-case starting point (SCENARIO_LINES.csv), not an external figure.